

Related Roadmaps

✓ AI Engineer Roadmap

✓ Machine Learning Roadmap

✓ C++ Roadmap

✓ Docker Roadmap

✓ Kubernetes Roadmap

Find the detailed version of this roadmap along with other similar roadmaps

roadmap.sh

Inference Engineer

Basic Terminology

Time to First Token

Tokens Per Second

Inter-Token Latency

Latency vs. Throughput

Latency Percentiles

End-to-End Metrics

Real-Time Factor

Latency Metrics

Prefill & Decode Phases

Online vs. Offline Inf.

Asynchronous Inference

Inference Phases

AI-Native Applications

Consumer vs. B2B

Reasoning Models

Application Context

Model Selection

Model Evaluation

Fine-Tuning

Distillation

Model Strategy

Compute

Memory

Instances & Interconnects

Sizing & Procurement

Desktop Inference

Mobile Inference

Kernel Selection & Fusion

safetensors

TensorRT

ONNX Runtime

AIPerf

Locust

SGLang Genai-bench

Performance Benchmarking

Quantization Approaches

Quantization-Aware Training

Post-Training Quantization

ModelOpt

Caching Approaches

KV Cache Storage

Cache-Aware Routing

Long Context Handling

NVIDIA Dynamo

Embedding Architectures

BERT-style models

LLM-based models

Image Generation Inference

Video Generation Inference

Dependency Management

NIMs

Kubernetes

GPU Procurement

Capacity Management

Geo-Aware Load Balancing

Security and Compliance

Zero-Downtime Deplo.

Cost Estimation

Observability

Introduction

Open vs Closed Models

Inference vs. Training

Shared vs. Dedicated Inf.

The Inference Stack

Models & Performance

Neural Networks

Encoder-Decoder Arch.

Machine Learning

Large Language Models

Transformer Blocks

Attention Mechanism

KV Cache

Mixture of Experts (MoE)

AI Engineer

Image & Video Models

Diffusion Models

Image Generation Arch.

Few-Step Image Generation

Video Generation Models

Bottleneck Analysis

Ops:Byte Ratio

Arithmetic Intensity

Roofline Model

LLM Bottlenecks

Image & Video Bottlenecks

Attention Optimization

FlashAttention

PagedAttention

Attention Variants

Hardware

GPU Architecture

Hopper

Blackwell

Grace & Vera

GPU Models

NVIDIA

Other Accelerators

Local & Edge Inference

Software

CUDA

Writing Custom Kernels

Frameworks & Libraries

PyTorch

Transformers & Diffusers

Model Formats & Runtimes

vLLM

SGLang

TensorRT-LLM

NVIDIA Dynamo

Inference Engines

Benchmarking & Profiling

Profiling Performance

Inference Techniques

Quantization

Measuring Quality Impact

Speculative Decoding

Speculation Algorithms

Draft-Target Decoding

EAGLE

N-gram

Caching

Model Parallelism

Pipeline Parallelism

Tensor Parallelism

Expert Parallelism

Multi-Node Inference

Disaggregation

Inference Beyond LLMs

Vision Language Models

Video Processing for VLMs

Omni-Modal Models

Embedding Models

Embedding Model Inference

Speech

Voice Activity Detection

Auto. Speech Recognition

Text-to-Speech (TTS)

Streaming TTS

Speech-to-Speech Models

Image & Video Generation

Production

Containerization

Docker

Autoscaling

Continuous Batching

Cold Starts

Routing & Load Balancing

LoRA-Aware Routing

Request Queueing

Scale to Zero

Indep. Component Scaling

Multi-Cloud

Testing & Deployment

Also visit the following related roadmaps

AI Engineer

Machine Learning

C++

Docker

Kubernetes